



## Semester 2 Examinations 2024–2025

**Course Instance Codes** 3BA1, 4BME1, 4BCT1, 4BMS2, 4BS2, 4FM2, 1GDS1, 1MDX1, 1SPS1  
**Exams** 3rd Arts, 4th Science, Masters, Structured PhD

**Module Code** CS4423

**Module** Networks

**Paper No** 1

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**Internal Examiner** Dr Niall Madden ✱

**Instructions:** Answer all 7 questions

**Duration** 2 Hours  
**Number of Pages** 4 (including this page)  
**Discipline** Mathematics

### Requirements

|                            |    |                                     |     |                                     |
|----------------------------|----|-------------------------------------|-----|-------------------------------------|
| Release in Exam Venue      | No | <input type="checkbox"/>            | Yes | <input checked="" type="checkbox"/> |
| Release to Library         | No | <input type="checkbox"/>            | Yes | <input checked="" type="checkbox"/> |
| Formulae and Tables        | No | <input type="checkbox"/>            | Yes | <input checked="" type="checkbox"/> |
| Handout                    | No | <input checked="" type="checkbox"/> | Yes | <input type="checkbox"/>            |
| Cambridge Tables           | No | <input checked="" type="checkbox"/> | Yes | <input type="checkbox"/>            |
| Graph Paper                | No | <input checked="" type="checkbox"/> | Yes | <input type="checkbox"/>            |
| Log Graph Paper            | No | <input checked="" type="checkbox"/> | Yes | <input type="checkbox"/>            |
| Graphic material in colour | No | <input checked="" type="checkbox"/> | Yes | <input type="checkbox"/>            |

### End of Requirements

Q1. [TOTAL: 14 MARKS]

- (a) [5 MARKS] Give an example (e.g., by sketching) of a simple connected graph of order 5, and size 8.

Is there any graph of order 5 and size 8 that is a tree? Explain your answer.

Explain why there is no simple connected graph of order 5 and size 12.

- (b) [4 MARKS] Consider the graph,  $G_1$ , shown in Figure 1. Write down the adjacency matrix,  $A_1$ , for  $G_1$ .

- (c) [5 MARKS] Is it possible to find a permutation matrix,  $P$ , such that  $P^T A_1 P$  has the form

$$P^T A_1 P = \begin{pmatrix} A_{11} & \mathbf{0} \\ \mathbf{0}^T & A_{22} \end{pmatrix}$$

where  $\mathbf{0}$  is a zero matrix, and  $A_{11}$  and  $A_{22}$  are square matrices? Either give such a matrix  $P$ , or explain clearly why this is not possible.

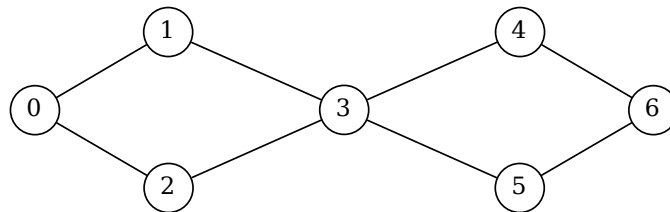


Figure 1: Graph  $G_1$  from Question 1

Q2. [TOTAL: 15 MARKS]

- (a) [5 MARKS] In Network Science, explain what an *affiliation network* is, giving an example of a scenario that is modelled by an affiliation network.

Why is the graph representation of an affiliation network necessarily bipartite?

- (b) [5 MARKS] Sketch the bipartite graph,  $G_2$ , on the nodes  $\{0, 1, \dots, 8\}$  with edges  $0 - 1$ ,  $0 - 4$ ,  $0 - 8$ ,  $2 - 3$ ,  $2 - 5$ ,  $3 - 4$ ,  $4 - 6$ ,  $5 - 7$ ,  $6 - 7$ ,  $6 - 8$ , and indicate a two-colouring.

- (c) [5 MARKS] Suppose  $G = (X, E)$  is a bipartite graph, with parts  $X_1$  and  $X_2$ . Explain what is meant by a *projection* of  $G$  on to  $X_1$ .

Recall  $G_2$  from Q2(b). Construct the projection of  $G_2$  onto the part that contains Node 8, and sketch the resulting graph.

Q3. [TOTAL: 15 MARKS] Consider the graph,  $G_3$ , generated by the following `networkx` instruction:

```
G3 = nx.Graph([[0,3], [1,3], [1,4], [1,6], [2,4], [2,5], [3,4],
               [3,6], [4,6]])
```

- (a) [3 MARKS] Sketch  $G_3$ .
  - (b) [3 MARKS] Calculate the *normalised degree centrality* of all nodes in  $G_3$ .
  - (c) [3 MARKS] Write down the *distance matrix* for this graph.  
Use that information to determine the *diameter* of  $G_3$ .
  - (d) [3 MARKS] Compute the *closeness centrality* of all nodes in  $G_3$ .
  - (e) [3 MARKS] Give an example of a graph for which a particular node would have the highest closeness centrality, but lowest degree centrality.
- .....

Q4. [TOTAL: 14 MARKS]

- (a) [5 MARKS] Describe Breadth First Search as an algorithm for computing distances between nodes in a (simple) graph. What is its input, what is its output, and what sequence of steps is taken to produce the output from the input?
- (b) [9 MARKS] Consider the graph,  $G_4$  shown in Figure 2. Show how to apply the Breadth First Search algorithm, starting at Node 0, to determine, for every node, its *predecessors* on the shortest path between it and Node 0.  
Use this information to list all shortest paths from Node 0 to Node 7.

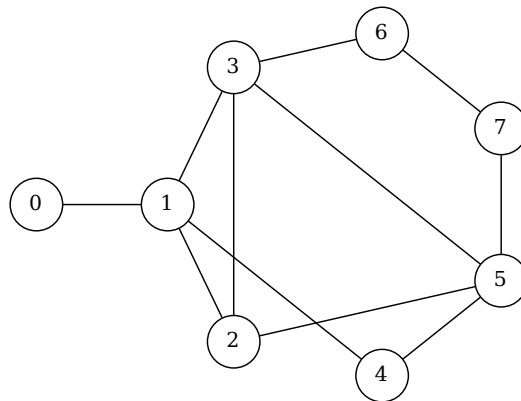


Figure 2: Graph  $G_4$  from Question 4

Q5. [TOTAL: 14 MARKS]

- (a) [2 MARKS] Let  $G_5$  be the tree on the nodes  $\{0, 1, 2, 3, 4, 5, 6\}$  that has as its *Laplacian matrix*

$$\begin{pmatrix} 3 & -1 & -1 & -1 & 0 & 0 & 0 \\ -1 & 3 & 0 & 0 & -1 & -1 & 0 \\ -1 & 0 & 2 & 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Sketch  $G_5$ .

- (b) [6 MARKS] How does one construct the Prüfer code for a tree?  
Compute the Prüfer code for  $G_5$  from Part (a).
- (c) [6 MARKS] How does a tree's Prüfer code relate to its degree sequence?  
Construct the degree sequence for the tree on the nodes  $\{0, 1, 2, 3, 4, 5\}$  with Prüfer code  $(2, 3, 2, 3)$ . Then construct and sketch the tree itself.
- .....

Q6. [TOTAL: 14 MARKS]

- (a) [4 MARKS] Define the two Erdős-Rényi models,  $G_{ER}(n, m)$  and  $G_{ER}(n, p)$  of random graphs. In these models, for what  $m$  and  $p$ , respectively, is the resulting graph  $K_n$ ?
- (b) [5 MARKS] In each model, what is the probability that a randomly chosen graph  $G$  has exactly  $m$  edges? Justify your answer.
- (c) [5 MARKS] A graph on 100 nodes is constructed by rolling a (fair) 6-sided die once for each possible edge: the edge is added only if the number shown is *even*. What is the probability that a node chosen at random has degree 45? (You do not need to compute a numerical value. It is enough to give an explicit formula in terms of the given data).
- .....

Q7. [TOTAL: 14 MARKS]

- (a) [4 MARKS] What is the *node clustering coefficient* of a node  $x$  in a graph  $G$ ? What is the graph clustering coefficient  $C$  of  $G$ ?
- (b) [5 MARKS] Determine the graph clustering coefficient  $C$  of a random graph in the  $G_{ER}(n, p)$  model. How does  $C$  behave in the limit  $n \rightarrow \infty$ , when the average node degree is kept constant? What practical consequence does this observation have?
- (c) [5 MARKS] Describe the *Watts-Strogatz small-world model* (WS model). What properties does a random graph sampled from the WS model have, that one wouldn't find in a random graph sampled from the  $G_{ER}(n, p)$  model, or in an  $(n, d)$ -circle graph?